

COURSE: AI ASSISTED CODING

Lab 7: Error Debugging with AI: Systematic approaches to finding

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Task 1 – Runtime Error Due to Invalid Input Type

- A Python program accepts user input and performs arithmetic operations. However, the program throws a runtime error because the input is treated as a string instead of a numeric type.

Example (Buggy Code):

```
num = input("Enter a number: ")
```

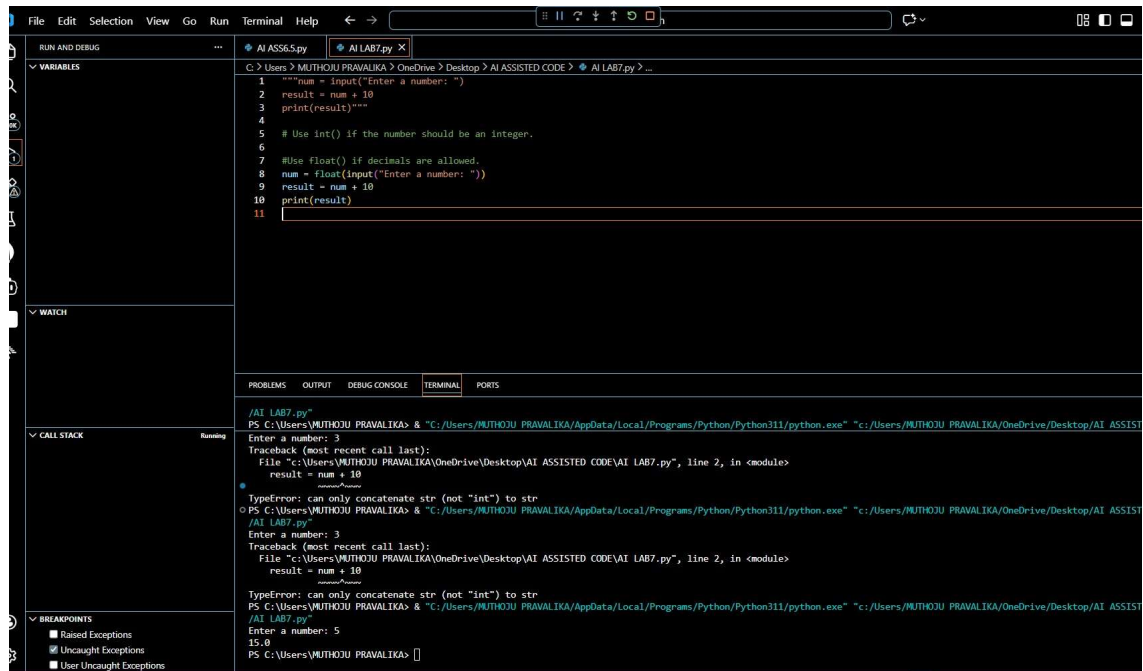
```
result = num + 10 print(result)
```

• Task:

Use AI tools to identify the cause of the runtime error and modify the program so it executes correctly.

Expected Output -1:

- AI converts the input to the appropriate numeric type and eliminates the runtime error.



Task Description

Task 2 – Incorrect Function Return Value

A function is designed to calculate the square of a number, but it does not return the computed result properly.

Example (Buggy Code):

```

def square(n): result =
n * n

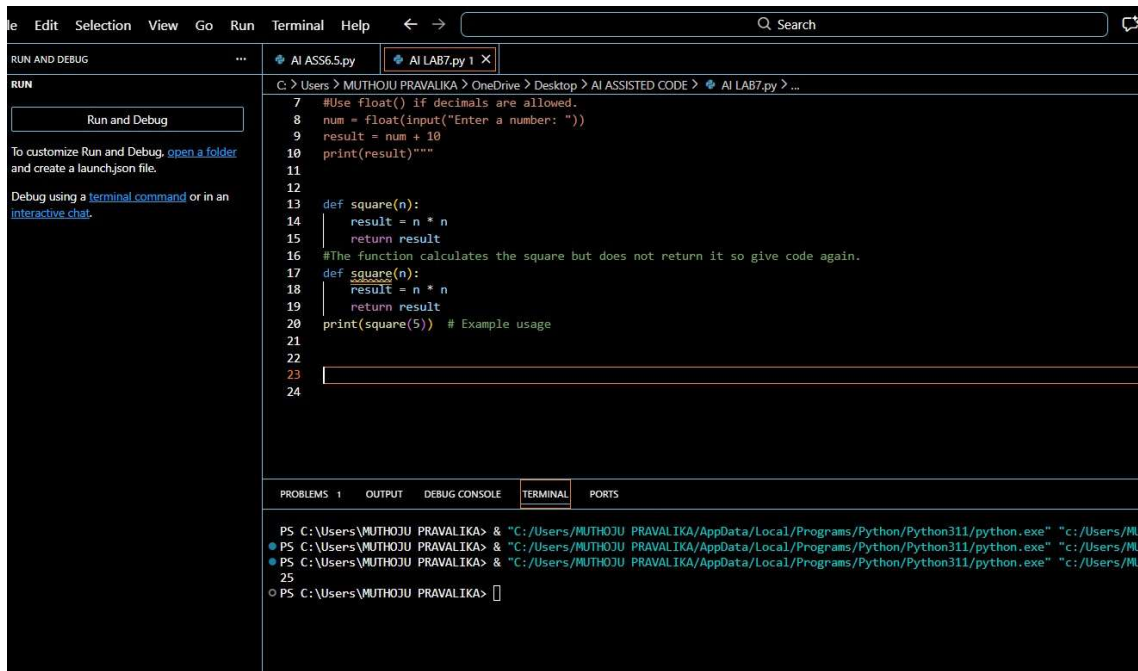
```

Task:

Use AI assistance to analyze the function and ensure the correct value is returned.

Expected Output -2:

AI fixes the missing return statement and the function returns the correct output.



Task Description

Task 3 – IndexError in List Traversal

A Python program iterates over a list using incorrect index limits, causing an IndexError. Example (Buggy Code): numbers = [10, 20, 30] for i in range(0, len(numbers)+1):

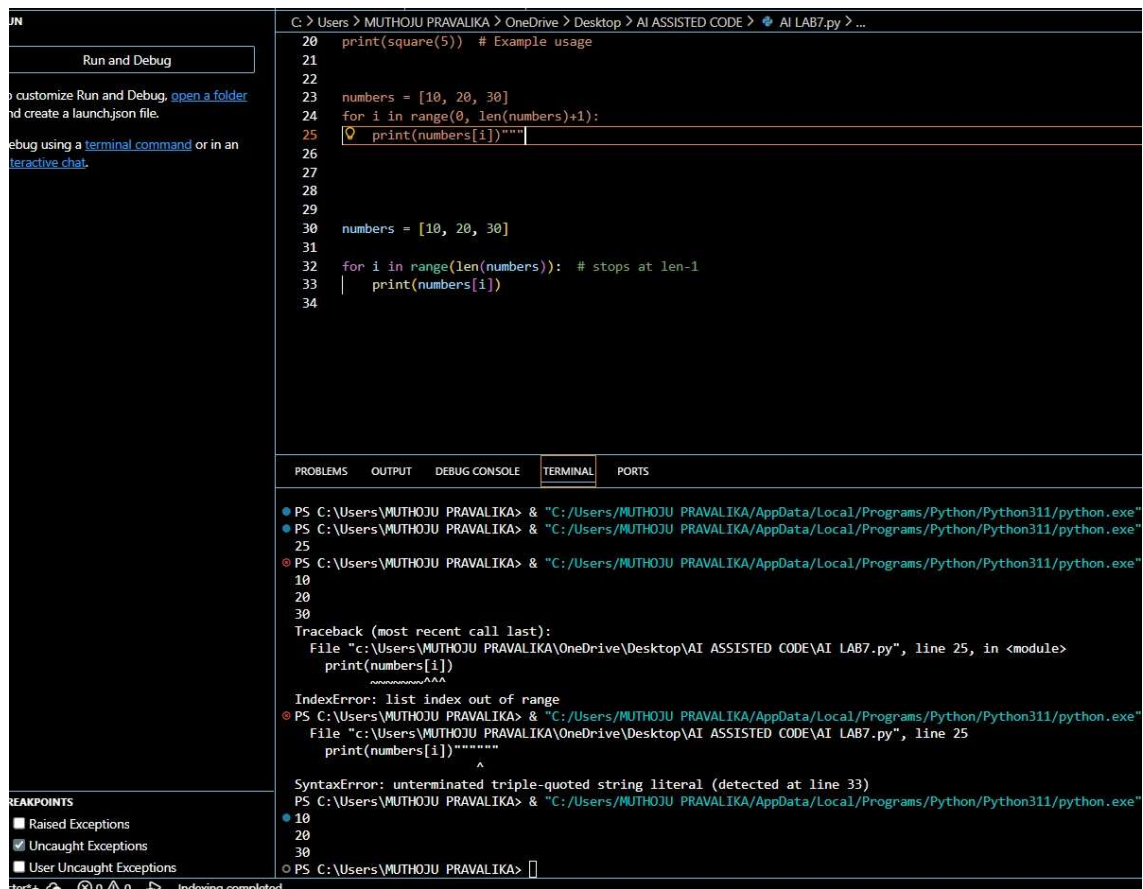
```
print(numbers[i])
```

Task:

Use AI to identify the incorrect loop boundary and correct the iteration logic.

Expected Output -3:

AI fixes the loop condition and prevents out-of-range list access.



Task 4 – Uninitialized Variable Usage

A program uses a variable in a calculation before assigning it any value.

Example (Buggy Code):

if True:

pass

print(total)

Task:

Use AI tools to detect the uninitialized variable and correct the program.

Expected Output -4:

AI initializes the variable correctly before it is used

```
31
32 for i in range(len(numbers)): # stops at len-1
33 |   print(numbers[i]) """
34
35
36 """if True:
37 |   pass
38 print(total)"""
39
40
41
42 # Task 4 - Uninitialized Variable Fix
43 total = 0 # Initialize the variable before use
44
45 # Example operation
46 if True:
47 |   total += 5
48
49 print("Total:", total)
50
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\MUTHOJU PRAVALIKA> & "C:/Users/MUTHOJU PRAVALIKA/AppData/Local/Programs/Python/Python
Traceback (most recent call last):
  File "c:\Users\MUTHOJU PRAVALIKA\OneDrive\Desktop\AI ASSISTED CODE\AI LAB7.py", line 38, in <m
    print(total)
    ^^^^^
NameError: name 'total' is not defined
PS C:\Users\MUTHOJU PRAVALIKA> & "C:/Users/MUTHOJU PRAVALIKA/AppData/Local/Programs/Python/Python
Total: 5
PS C:\Users\MUTHOJU PRAVALIKA>
```

Task Description

Task 5 – Logical Error in Student Grading System

A grading program assigns incorrect grades due to improper conditional logic.

Example (Buggy Code):

```
marks = 85
if marks >= 90:
    grade = "A"
elif marks >= 80:
    grade = "C"
else:
    grade = "B"
print(grade)
```

Task:

Use AI to analyze the grading conditions and correct the logical flow.

Expected Output -5:

AI corrects the conditional logic so grades are assigned accurately.

The screenshot shows the Visual Studio Code interface with a Python file named 'AI LAB7.py' open. The editor displays a Python script for assigning grades based on marks. The script includes a comment '# Task 5 - Logical Error Fix in Grading System' and a line 'marks = int(input("Enter marks: ")) # Convert input to integer'. The conditional logic for grade assignment is as follows:

```
58 elif marks >= 80:
59     grade = "C"
60 else:
61     grade = "B"
62 print(grade)
63 """
64
65
66 # Task 5 - Logical Error Fix in Grading System
67 marks = int(input("Enter marks: ")) # Convert input to integer
68
69 if marks >= 90:
70     grade = "A"
71 elif marks >= 80:
72     grade = "B"
73 elif marks >= 70:
74     grade = "C"
75 else:
76     grade = "D"
77
78 print("Grade:", grade)
79
```

The 'RUN AND DEBUG' sidebar on the left shows the 'RUN' button and instructions: 'To customize Run and Debug, [open a folder](#) and create a launchjson file.' and 'Debug using a [terminal command](#) or in an [interactive chat](#).'

The 'TERMINAL' tab at the bottom shows the execution output:

```
PS C:\Users\MUTHOJU PRAVALIKA> & "C:/Users/MUTHOJU PRAVALIKA/AppData/Local/Programs/Python/Python311/python.
C
PS C:\Users\MUTHOJU PRAVALIKA> & "C:/Users/MUTHOJU PRAVALIKA/AppData/Local/Programs/Python/Python311/python.
Enter marks: 98
Grade: A
PS C:\Users\MUTHOJU PRAVALIKA>
```